

Multimodal Learning for Semantic-Aware Hateful Meme Detection in Low-Resource Kannada Memes

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Abstract—Detecting hateful content in Kannada memes is challenging given the scarcity of labeled data and the multimodal nature of the content. This paper presents a multimodal framework, tested on a small dataset of 140 memes with OCR-extracted text. We benchmarked various machine learning and deep learning architectures, including fusion techniques involving IndicBERT, mBERT, and CNN. The most effective configuration was a multimodal fusion model that leveraged a language-specific IndicBERT for language-specific textual features and a CNN for visual analysis, achieving an F1-score of 0.81 and an accuracy of 71%. We incorporate a knowledge graph that maps hateful keywords to underlying semantic topics, providing essential interpretability. This work establishes a robust, effective and interpretable baseline for hate speech detection within the underrepresented domain of Indic language meme analysis.

Index Terms—Hateful memes, Multimodal learning, Meme analysis, Image-text fusion, Code-mixing, Low-resource NLP, Early fusion, IndicBERT, mBERT, knowledge graph, Convolutional neural network (CNN), CLIP

I. INTRODUCTION

Memes have emerged as a dominant medium of expression on the internet. Despite their humorous appeal, memes can be potent carriers of hate, sexism, casteism, and other forms of online abuse. While hateful meme detection in English has been extensively studied [6] [13], the Kannada language—a major Indian language with over 50 million speakers—remains underexplored due to limited NLP resources. We address this gap through a multimodal pipeline that combines text and image understanding, enriched with knowledge graph reasoning.

The contributions of the study include:

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- Develop a curated list of Kannada hate speech keywords, and categorize them into semantic topics (e.g., gender abuse, politics, caste-based hate).
- Implement and evaluate machine learning and deep learning models (SVM, Random Forest, Logistic Regression, IndicBERT [5], mBERT [3], CNN) for text-only and image-only meme classification.
- Construct a knowledge graph linking meme texts, hate keywords, semantic topics, and labels to provide interpretable hate reasoning.

II. RELATED WORK

Research into the automatic detection of hateful content has evolved significantly, particularly with the rise of multimodal communication platforms like social media. Our work builds upon several key areas of this research, including multimodal hate detection, Indian language NLP, and the application of advanced fusion and knowledge representation techniques.

A. Multimodal Hate Detection

The importance of combining visual and textual cues for hate speech detection was underscored by seminal works such as The Hateful Memes Challenge [6], which provided a benchmark dataset to evaluate models on this challenging task. Similarly, the MultiOFF dataset [13] further advanced the field by offering a dataset specifically designed for multimodal offensive content. These foundational efforts established that models must be able to understand the interplay between text and image to accurately identify harmful content, a principle that guides our fusion-based approach.

B. Indian Language Models and Datasets

Recognizing that most research has focused on English, several studies have addressed the unique challenges of of-

fensive content in Indian languages. A benchmark dataset for code-mixed Dravidian languages (DravidianCodeMix) [2] expanded resources for offensive language detection, while subsequent efforts such as HASTIKA [10] specifically targeted Kannada-English hate speech and the work by Hegde et al. [4] introduced critical datasets for offensive language in Dravidian and Kannada languages, respectively. To address the linguistic diversity, powerful transformer models have been developed, such as IndicBERT [5], which is pre-trained on a vast corpus of Indian languages, and the specialized Krishiq-BERT [1]. These models serve as the basis for our textual feature extraction, providing a strong foundation for our language-specific analysis.

C. Fusion and Knowledge-Based Techniques

Beyond basic fusion, researchers have explored more sophisticated methods to combine information and add interpretability. Dual-stage fusion techniques have been proposed by works like MemeFier [8] and MEMEX [11], which process each modality separately before a second-stage fusion. Other studies, such as KnowMeme [9] and MemeGraphs [7] have demonstrated the value of graph reasoning for meme analysis. Some approaches, such as IMKG [14], enrich meme analysis by linking external knowledge sources (e.g., Wikidata), which inspired our decision to adopt a simplified knowledge graph design. While our work uses a simpler fusion approach, it leverages the power of a custom-built knowledge graph to provide similar interpretability, mapping hate keywords to semantic concepts in a manner similar to these advanced techniques.

Our work falls within the broader context of hate detection surveys and the application of novel techniques like federated learning [12], contributing a focused study on a low-resource language to the growing body of literature on this critical societal issue.

III. METHODOLOGY

We began by procuring the Kannada memes sourced from Instagram. The workflow illustrating the steps from data collection through classification and knowledge graph mapping is summarized in Figure 1.

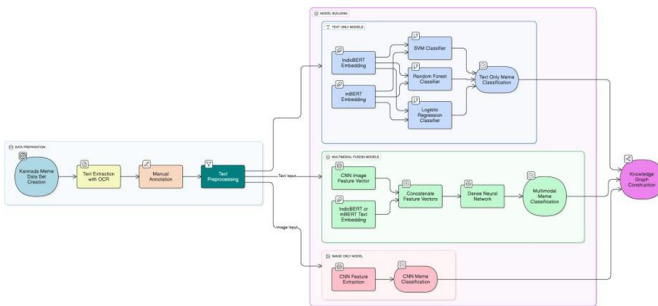


Fig. 1. Approach for Meme Classification using Multimodal Approach

A. Text Acquisition and Corpus Building

1) *Meme Collection*: We collected 140 Kannada memes from Instagram via screenshots saved as .jpg files. All meta-data and labels were stored in meme_extracted.xlsx.

2) *Text Extraction (OCR)*: We used PyTesseract¹ with lang='kan'² to extract text. Roughly 15% of the memes had OCR failures, for which Google Lens was used manually.

3) *Manual Annotation*: Each meme was annotated as either 'Hateful' or 'Non-hateful' based on the text and image context. A meme is labelled as 'Non-hateful' if it is a funny meme or a meme that elicits humour strictly, otherwise, it is labelled as 'Hateful'. Table I instantiates such memes in Kannada.

TABLE I
SAMPLE KANNADA MEMES

Non-hateful	Hateful
RCB 8 ನೇ ಹ್ಯಾಪ್ಪಿನಾಗಿ ರಜತ್ ಆಯ್ಕೆಯಾಗಿದ್ದಾರೆ 	ಕತ್ತಲ ಕಾಮುಕ ಕೆಲಸದಲ್ಲಿ ಲಾಕ್! 

B. Pre-processing

Non-Kannada characters in memes were removed. Standard preprocessing pipeline involving tokenization, stopword removal, and lowercasing was applied.

C. Meme Classification using Text-Only Models

To classify memes based exclusively on their textual content, we developed a two-stage methodology: contextual embedding extraction and subsequent classification using traditional machine learning models. This approach leverages the power of large-scale pre-trained language models to generate rich semantic representations, which are then used as input for simpler, yet effective, classifiers.

1) *Contextual Embedding Extraction*: The first stage involved transforming the raw text of the memes into dense vector representations. This process, known as embedding extraction, is crucial for capturing the semantic and syntactic nuances of the text. We utilized two distinct pre-trained transformer-based models to generate these embeddings:

- **IndicBERT**: This model was selected for its specific training on a large corpus of Indian languages, making it particularly adept at handling the linguistic diversity and code-switching commonly found in Indian memes.
- **mBERT (multilingual BERT)**: As a model pre-trained on a vast array of languages, mBERT provides a robust baseline for capturing multilingual patterns, which is essential

¹<https://pypi.org/project/pytesseract/>

for memes that may incorporate English, regional Indian languages, or a mix of both.

After tokenizing the cleaned meme texts, each token sequence was passed through both IndicBERT and mBERT. The output of these models provides a fixed-size vector representation for each text. Specifically, we extracted 768-dimensional contextual embeddings, which are dense numerical representations that encode the meaning of the words in their specific context within the meme text. These embeddings serve as feature vectors for the subsequent classification task.

2) *Classification using Traditional Machine Learning Models*: The extracted 768-dimensional embeddings were used to train and evaluate three different machine learning classifiers. The purpose of this evaluation was to determine which model could most effectively learn the relationship between the contextual embeddings and the target labels (hateful or non-hateful). The selected models were:

- Logistic Regression: A linear model that estimates the probability of a binary outcome. Its simplicity and interpretability make it an effective baseline for binary classification tasks.
- Random Forest: An ensemble learning method that constructs a multitude of decision trees during training. It is known for its ability to handle non-linear relationships and its robustness to overfitting.
- Support Vector Machine (SVM): A powerful discriminative classifier that finds the optimal hyperplane to separate the data points into different classes. SVMs are particularly effective in high-dimensional spaces, making them well-suited for our 768-dimensional embeddings.

D. Meme Classification using Image-only Models

To classify memes using visual features alone, we built a Convolutional Neural Network (CNN) model. The CNN model is designed for binary classification and is structured to progressively extract and refine features from the input data. The architecture consisted of a sequence of convolutional blocks, a flattening layer, and a fully connected (dense) classifier.

The initial layers of the network are dedicated to hierarchical feature extraction. The input data is processed by two consecutive convolutional blocks, each comprising a Conv2D layer followed by a rectified linear unit (ReLU) activation and a MaxPooling2D layer.

The features extracted by the convolutional blocks are then prepared for classification. The output from the final MaxPooling2D layer, which is a multi-dimensional feature map, is flattened into a one-dimensional vector. The flattened feature vector is fed into a dense fully connected layer. To combat overfitting, a Dropout layer is applied. During training, Dropout randomly sets a fraction of input units to zero at each update.

The final layer is a single-neuron dense layer with a sigmoid activation function. The sigmoid function, given by $\sigma(x) = \frac{1}{1+e^{-x}}$, squashes the output value to a range between 0 and 1, interpreted as a probability. The overall architecture combines feature extraction via convolution and pooling with a dense

classifier, a common and effective approach for image and pattern recognition tasks.

E. Meme Classification using Multimodal Fusion Model

To effectively leverage both the visual and textual components of memes, we developed a multimodal fusion model. The 'early fusion' approach we adopted combines the feature representations of different modalities at an early stage of the model architecture, allowing the network to learn joint representations that capture the interplay between text and image. Our fusion models consisted of two distinct branches—a convolutional neural network (CNN) for processing images and a transformer-based model for processing text—which are then combined for a unified classification task.

The architecture of our early fusion model is structured as follows:

1) *Image Feature Extraction*:: The meme image is processed through a dedicated CNN. This CNN is designed to hierarchically extract and learn rich visual features from the image. The final layer of the CNN produces a dense 512-dimensional feature vector, which serves as a compact representation of the visual content. Image features were extracted using a pretrained ResNet-50 model. The images were resized to 128x128 and normalized using standard ImageNet parameters.

2) *Text Feature Extraction*:: Concurrently, the meme's textual content is processed through a transformer-based model, either IndicBERT or mBERT. These pre-trained models generate a 768-dimensional contextual embedding that encapsulates the semantic meaning of the text. The choice between IndicBERT and mBERT allows us to evaluate the impact of language-specific versus multilingual text representations on multimodal performance.

3) *Feature Concatenation and Classification*:: The two feature vectors—the 512-dimensional visual vector from ResNet-50 and the 768-dimensional textual vector from the transformer—are concatenated into a single, combined feature vector. This concatenation step creates a joint representation that fuses the information from both modalities. This fused vector is then passed to a fully connected dense neural network. To prevent overfitting and enhance the model's generalization capabilities, a dropout regularization layer is applied within this dense network. The final layer of the network is a single neuron with a sigmoid activation function, which outputs a probability score for the binary classification task (hateful vs. non-hateful).

F. Building a Knowledge Graph for Hate Speech Detection

To analyze the relationship between hate keywords, their semantic categories, and individual memes, we constructed a knowledge graph. This approach allowed us to move beyond a simple keyword-matching system by visualizing and modeling the complex connections within our dataset.

1) *Graph Construction*: Using the networkx² library, we built a comprehensive graph with four types of nodes:

²<https://networkx.org/>

- **Text Nodes:** Each individual meme text in our dataset was represented as a unique node. These nodes are the core data points of our analysis.
- **Keyword Nodes:** We created a curated list of approximately 35 Kannada hate keywords
- **Topic Nodes:** To provide a deeper semantic context, each hate keyword was linked to a specific semantic category or "topic." Examples of these topics include gender, politics, and vulgarity. Each of these topics also had its own node.
- **Label Nodes:** We added two final nodes representing the classification labels for our task: Hateful and Non-Hateful.
- **Defining the Connections (Edges):** The connections, or edges, between these nodes represented specific relationships within the data:
 - *contains_keyword:* An edge was drawn from a Text Node (a meme) to a Keyword Node if the meme text contained that specific hate keyword.
 - *targets_topic:* An edge was drawn from a Keyword Node to a Topic Node to show which semantic category that keyword belonged to.
 - *has_label:* An edge was drawn from a Text Node (a meme) to a Label Node to indicate its final classification as either Hateful or Non-Hateful.

IV. EXPERIMENTAL SETUP AND EVALUATION

A. Dataset Characteristics and Considerations

As was elaborated in Section III-A the dataset used for this study was a collection of 140 meme images, each paired with its corresponding textual content. The characteristics and inherent challenges of this dataset are detailed in the subsections that follow.

1) *Size and Structure:* The dataset comprises 140 unique meme images. The accompanying text for each image is of varying length, reflecting the natural diversity of digital communication.

2) *Language:* The primary language of the textual content is Kannada. However, the dataset also features instances of English and a significant amount of code-mixing, a common phenomenon in digital communication where speakers switch between two or more languages within a single conversation. This linguistic complexity necessitates the use of robust models like IndicBERT and mBERT, which are designed to handle multilingual and code-mixed content.

3) *Bias and Generalizability:* The data was sourced from social platforms, which introduced a natural skew towards culturally sensitive or provocative themes. This inherent bias is an important consideration, as it may affect the generalizability of the trained models to memes from different cultural contexts or with more neutral content. The models' performance should be interpreted within this specific domain.

B. Training Setup and Hyperparameters for Image-only Model

To ensure a robust and reproducible training process for our models, a consistent set of configurations and hyperparameters was used. The training setup was as follows:

1) *Input Image Size:* All meme images were resized to a uniform dimension of 128x128 pixels. This standardization is crucial for ensuring that the input to the convolutional neural network (CNN) branch is consistent and of a manageable size, balancing computational efficiency with sufficient detail retention.

2) *Batch Size:* A batch size of 16 was used during training. This hyperparameter determines the number of samples processed before the model's internal parameters are updated. A batch size of 16 was chosen to strike a balance between computational efficiency and stable gradient updates.

3) *Optimizer:* The Adam optimizer was employed for model training. Adam (Adaptive Moment Estimation) is an advanced optimization algorithm that combines the benefits of two other extensions of stochastic gradient descent: RMSprop and AdaGrad. It is computationally efficient, requires little memory, and well-suited for a wide range of deep learning problems.

4) *Loss Function:* For classifying memes as hateful vs. non-hateful, the Binary Cross-Entropy loss function (Equation 1) was used. The function measures the performance of a classification model whose output is a probability between 0 and 1.

$$L(y, \hat{y}) = -\frac{1}{N} \sum_{i=1}^N [y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)] \quad (1)$$

where N is the number of observations, y_i is the actual binary label of the i^{th} observation, and $\hat{y}$ is the predicted probability. Minimizing this function during training encourages the model to output a probability that is close to the true label.

C. CLIP-Based Multimodal Feature Extraction as Baseline

To establish a comparative baseline, we implemented a multimodal classification model leveraging features from CLIP (Contrastive Language-Image Pre-training). The experiment utilized the openai/clip-vit-base-patch32 model on our Kannada meme dataset (140 samples). We treated the full CLIP model as a static feature extractor, freezing all its parameters (image and text encoders) to prevent weight updates during training. For each meme, the 512-dimensional image features and 512-dimensional text features were extracted and concatenated to form a 1024-dimensional input vector. A small, custom feed-forward neural network (MultiModalClassifier) was trained on these features for the binary classification task. This classifier consisted of an input layer, a hidden layer (512 units, ReLU, and 30% dropout), and a 2-output layer. Due to the small dataset size (112 train, 28 test) and potential class imbalance, we applied balanced class weights and label smoothing ($\alpha = 0.1$) to the CrossEntropyLoss function. The model was trained for 20 epochs using the Adam optimizer with a learning rate of 1×10^{-4} . The training process was

stable, with the average batch loss steadily decreasing from 0.7045 (Epoch 1) to 0.3492 (Epoch 20).

D. Optimization of Fusion Models

To train and optimize our multimodal fusion models, a specific set of hyperparameters and techniques was employed to ensure efficient convergence and prevent overfitting. The optimization strategy is summarized in the subsections that follow.

1) *Optimizer*: We utilized the Adam optimizer, a popular and robust choice for training deep neural networks. Adam was configured with a specific learning rate of 0.0001. This low learning rate was chosen to allow for fine-tuned weight updates, helping the model converge to a stable minimum without overshooting the optimal solution, which is particularly important for complex multimodal architectures.

2) *Batch Size*: The batch size was set to 8. This smaller batch size, compared to the textual model, was selected to accommodate the increased memory requirements of processing both image and text data simultaneously. A smaller batch size can also provide a more regularized training process by introducing more noise into the gradient updates, which can sometimes lead to better generalization.

3) *Early Stopping*: To prevent the model from overfitting to the training data, an early stopping mechanism was implemented. This technique monitors the model's performance on a held-out validation set during training. If the validation loss fails to improve for a predefined number of epochs (patience), the training process is automatically stopped. This ensures that the final model is the one that performed best on unseen data, thereby improving its generalization capability.

4) *Performance Evaluation*: The performance of each classifier was rigorously evaluated using standard classification metrics. These metrics are crucial for providing a comprehensive view of model effectiveness, especially in an imbalanced dataset context. A robust assessment of our models, providing insights into their ability to accurately identify hateful content while minimizing misclassifications was carried out using the metrics below:

- **Precision (P)**: The ratio of correctly predicted positive observations to the total predicted positive observations. It measures the model's ability to avoid false positives.
- **Recall (R)**: The ratio of correctly predicted positive observations to all actual positive observations. It measures the model's ability to find all positive instances.
- **F-Measure (F)**: The harmonic mean of Precision and Recall. This metric provides a balanced assessment of the model's performance, particularly when there is an uneven class distribution.
- **Accuracy (A)**: The proportion of correctly classified instances out of the total number of instances.

V. RESULTS AND DISCUSSION

In the subsections that follow, the performance of text-only, image-only and multimodal fusion models are presented and discussed. Further, we present knowledge graphs for

visualizing hateful memes by mapping keywords in memes to semantic topics such as gender, caste, vulgarity and so forth.

A. Performance of the CLIP-Based Baseline

The CLIP feature-based classifier achieved an overall accuracy of 54% on the test set. The model demonstrated a notable disparity in performance between the two classes. The model performed better on Class 0 ('non-hateful') with an F1-score of 0.63. Performance on Class 1 ('hateful') was significantly weaker, yielding an F1-score of only 0.38. This outcome suggests that relying on static, pre-trained CLIP features alone - especially when the model was primarily trained on English data - presents a significant challenge for accurate classification on this small, low-resource Kannada meme dataset.

B. Performance of Various Classifiers with IndicBERT and mBERT Embeddings using Text-only Features

The task of classifying nuanced human emotion proved to be highly challenging for all models, highlighting the difficulty of predicting nuanced human emotion from text. Table II summarizes the performance of three different classifier models on the test set.

C. Performance of Image-only Model

We obtained an accuracy of only 43% while training a simple CNN model consisting of two Conv2D + Maxpool layers followed by dense layers using only image features. The results are as shown in Table III

D. Performance of Multimodal Fusion Model

Table IV shows the performance of multimodal classification approach employing textual and image features.

1) *mBERT based Text Embedding with CNN* : Fusion of 768-dim text embeddings (mBERT) and 512-dim CNN image embeddings yielded an accuracy of 64% and the model did not fare well in classifying hateful memes.

2) *IndicBERT based Text Embedding with CNN* : Fusion of 768-dim text embeddings (IndicBERT) and 512-dim CNN image embeddings enabled best results with 71% accuracy, using dropout and early stopping. However, mBERT with CNN failed in detecting hateful memes.

E. Knowledge Graph for Hateful Memes Visualization

To visualize the hateful memes, we created interactive knowledge graphs³ using vis.js in web browsers. Figures 2 through to 6 depicts the relationship between hate keywords, their semantic categories, and individual memes. The visualization effectively demonstrated how specific keywords and their underlying topics contributed to a meme's final classification.

F. Discussion

The development and evaluation of our models encountered several significant challenges and limitations, which are important to contextualize the results of our study.

³<https://knowledge-graph-kannada.netlify.app/>

TABLE II
PERFORMANCE OF VARIOUS CLASSIFICATION MODELS WITH INDICBERT AND mBERT EMBEDDINGS

Embedding	Classifier	Class	Precision	Recall	F1-Score	Accuracy
IndicBERT	SVM	Non-hateful	0.69	0.50	0.58	0.54
		Hateful	0.40	0.60	0.48	
	Random Forest	Non-hateful	0.65	0.83	0.73	0.61
		Hateful	0.40	0.20	0.27	
	Logistic Regression	Non-hateful	0.65	0.61	0.63	0.54
		Hateful	0.36	0.40	0.38	
mBERT	SVM	Non-hateful	0.70	0.78	0.74	0.64
		Hateful	0.50	0.40	0.44	
	Random Forest	Non-hateful	0.67	1.00	0.18	0.68
		Hateful	1.00	0.10	0.80	
	Logistic Regression	Non-hateful	0.71	0.67	0.69	0.61
		Hateful	0.45	0.50	0.48	

TABLE III
PERFORMANCE OF CNN MODEL USING IMAGE-ONLY FEATURES

Class	Precision	Recall	F1-Score	Accuracy
Non-hateful	0.64	0.37	0.47	0.43
Hateful	0.29	0.56	0.38	

TABLE IV
PERFORMANCE OF CNN MODEL USING MULTIMODAL FUSION

Embedding	Class	Precision	Recall	F1-Score	Accuracy
IndicBERT	Non-hateful	0.71	0.94	0.81	0.71
	Hateful	0.75	0.30	0.43	
mBERT	Non-hateful	0.64	1.00	0.78	0.64
	Hateful	0	0	0	

1) *Optical Character Recognition (OCR) Limitations*: A primary challenge was the unreliable performance of standard OCR tools on the meme images. The decorative and non-standard fonts commonly used in Kannada memes, particularly those with stylized layouts or artistic effects, often led to incorrect or incomplete text extraction. This limitation necessitated a careful manual review and cleaning process, as the quality of the textual input directly impacts the performance of the text-based models.

2) *Labor-Intensive Manual Annotation*: Creating the ground truth labels for our dataset was a highly labor-intensive process. Human annotators were required to carefully review each meme and its context to classify it as ‘hateful’ or ‘non-hateful’. This manual effort was not only time-consuming but also introduced a potential for subjectivity and inter-annotator disagreement, underscoring the complexity of defining and identifying hate speech.

3) *Small Dataset Size*: The limited size of our dataset consisting of 140 memes posed a significant constraint. While useful for an initial study, this small scale inherently restricts the models’ ability to generalize to a wider variety of meme content. Deep learning models, in particular, often require large amounts of data to learn robust, generalizable features, and our small dataset may lead to models that are prone to

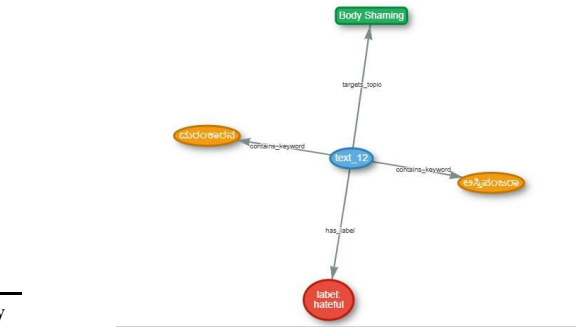


Fig. 2. Entity Mapping for the Topic Body Shaming

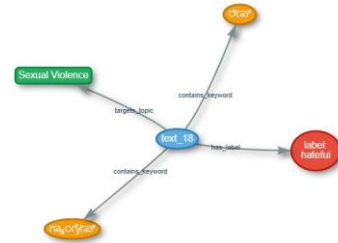


Fig. 3. Entity Mapping for the Topic Sexual Violence

overfitting and do not perform as well on unseen data.

4) *Keyword Engineering for the Knowledge Graph*: The construction of the knowledge graph, while insightful, relied on a manually curated list of hate keywords. This manual engineering is a potential source of bias, as the graph’s scope is limited by our predefined list of 35 Kannada words. This approach may fail to capture new or evolving hate speech terminology and nuances, highlighting a limitation in its ability to adapt to linguistic changes.

VI. CONCLUSION AND FUTURE WORK

This study introduces a preliminary but promising multimodal framework for the detection of hateful memes in Kannada, integrating both textual and visual analysis. Our methodology demonstrates the utility of combining traditional

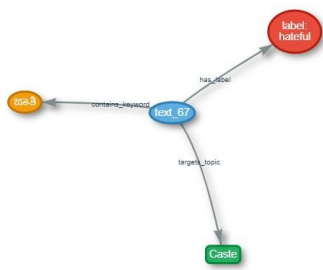


Fig. 4. Entity Mapping for the Topic Caste

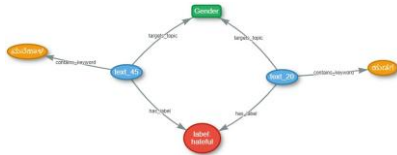


Fig. 5. Entity Mapping for the Topic Gender

NLP techniques, like knowledge graph construction, with advanced deep learning models.

The most effective model observed in this study was the IndicBERT + CNN fusion model. This finding highlights the importance of leveraging language-specific models (IndicBERT) to capture the subtleties inherent in code-mixed and regional content, especially in comparison to unimodal approaches. Furthermore, the construction of a knowledge graph provided a valuable post-hoc layer of interpretability. It served as a human-readable artifact, visually mapping the connections between specific hate keywords, their semantic topics, and the final classification, thereby offering insights into the underlying reasons for the model's classification. Crucially, this knowledge graph was not used as an input feature during model training and did not directly contribute to the reported quantitative performance metrics of the IndicBERT + CNN fusion model. It is important to emphasize that these conclusions are preliminary and should not be overgeneralized given the limited size of the current dataset. The risk of overfitting on the current data necessitates a cautious interpretation of the reported performance metrics.

Based on the challenges and findings of this study, several avenues for future research are identified to improve the robustness and scalability of the hateful meme detection framework. A significant increase in the dataset size is essential to train models that can generalize effectively beyond the current samples and mitigate the observed overfitting risk. Fine-tuning an OCR model specifically for non-standard Kannada fonts and crowd-sourced annotations would reduce reliance on manual annotations. Future work will focus on automatically enriching the knowledge graph with external sources, such as Wikidata, to expand the semantic network and allow the system to dynamically adapt to new hate topics and linguistic expressions. To enhance the robustness and

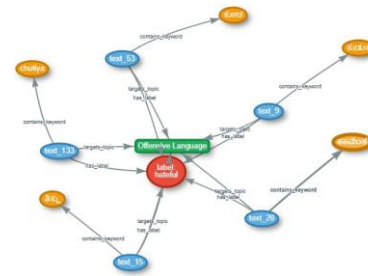


Fig. 6. Entity Mapping for the Topic Offensive Language

quantitative capabilities of our framework, a critical future step is the transition of the Knowledge Graph from a post-hoc tool to a structured feature component. In summary, this study provides a foundational blueprint and encouraging initial evidence for effective multimodal hate speech detection in Kannada, establishing a clear agenda for future research to achieve robustness and scalability.

REFERENCES

- [1] P. Ajawan, P. Kumar, and M. Singh, "Krishiq-BERT: A Pre-trained Transformer Model for Agricultural Texts in Indian Languages," *Journal of Information Engineering and Applications, Series B*, 2024.
- [2] B. R. Chakravarthi, R. Priya, S. Sangeetha, K. Suganya, and M. Vijayakumar, "DravidianCodeMix: A Benchmark Dataset for Code-Mixed Social Media Text in Dravidian Languages," in *Proceedings of the 13th Language Resources and Evaluation Conference (LREC)*, 2022.
- [3] J. Devlin, M. Chang, K. Lee, and K. Toutanova, "BERT: pre-training of deep bidirectional transformers for language understanding," *CoRR*, vol. abs/1810.04805, 2018. [Online]. Available: <http://arxiv.org/abs/181004805>.
- [4] A. Hegde and S. Lakshmaiah, "Trolling Memes in Kannada: Dataset and Analysis of Multimodal Offensive Content," *arXiv preprint arXiv:2403.01234*, 2024.
- [5] D. Kakwani, A. Panyam, A. Goyal, and A. Goyal, "IndicNLP Suite: Pre-training a Transformer for Indian Languages," *arXiv preprint arXiv:2005.00085*, 2020.
- [6] D. Kiela, A. Firoj, A. Singh, Y. Soni, H. Jhamtani, M. Singh, and Y. Chen, "The Hateful Memes Challenge: A Contest to Advance Multimodal Hate Speech Detection," *arXiv:2005.00085*, 2020.
- [7] V. Kougia, S. Fetzl, T. Kirchmair, E. C. ano, S. M. Baharlou, S. Sharifzadeh, and B. Roth, "Memegraphs: Linking memes to knowledge graphs," 2023. [Online]. Available: <https://arxiv.org/abs/230518391>.
- [8] C. Koutlis, S. Psarrou, and A. Vlachos, "MemeFier: A Two-Stage Multimodal Fusion Model for Hate Speech Detection in Memes," *arXiv preprint arXiv:2304.02906*, 2023.
- [9] L. Shang, Y. Li, Q. Zhang, and M. Huang, "KnowMeme: A Knowledge-Enhanced Approach for Multimodal Hateful Meme Detection," in *Proceedings of the 17th International Conference on e-Science*, 2021.
- [10] A. Sharma, A. Kumar, and A. Singh, "HASTIKA: A Hate Speech Corpus for Kannada and English Code-Mixed Social Media Content," *Language Resources and Evaluation*, 2025.
- [11] S. Sharma, S. Garg, S. Singh, and A. Sharma, "MEMEX: Multimodal Extreme Learning for Hateful Meme Detection," in *Proceedings of the 61st Annual Meeting of the ACL*, 2023.
- [12] A. Singh and R. Thakur, "Federated Hate Detection: A Privacy-Preserving Approach for Hateful Meme Identification," in *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics (NAACL)*, 2024.
- [13] S. Suryawanshi, A. Sharma, A. Kumar, and A. Jindal, "MultiOFF Dataset: Multimodal Offensive Content Identification in Social Media," *Proceedings of the 12th LREC*, 2020.
- [14] L. Tommasini, L. Gatti, M. Tani, and A. Tisi, "IMKG: An Image-Meme Knowledge Graph for Multimodal Analysis," in *Graph-Based Representation and Reasoning*. Springer, 2023.